Additional Notes on the Meaning of Field and Potential

Price per unit stuff

> SYNOPSIS
>
> Here are some good analogies for field (for force) and potential (for energy).

The supermarket

You go to a supermarket and find bags of coffee beans. One bag costs **$12 for 8 ounces** and another **$15 for 12 ounces**. You just want the cheapest deal. Which gets you the most bang for the buck? How would you figure this out? Show your work below.

Now, think of force in newtons (N) or energy in joules (J) as the **total prices in dollars**. The most-bang-for-the-buck math you just did is the **field** (N/kg for gravitational or N/C for electrostatics) or **potential** (J/kg for gravitational or J/C for electrostatic): **the price per unit "stuff."**

One thing to straighten out before going further, because it is where this analogy usually misleads. A price per ounce sounds like a property of **the bag you carry home**. It is not — it is a property of **the store**. The same beans cost more across town, and what you pay is (your ounces) × (that store's price per ounce).

Field and potential work the same way round. **They belong to the place, not to the charge you bring.** The field at a point is set by whatever charges are making it, and it is sitting there at that value before you put anything in it — just as the store's price is on the shelf before you decide how much to buy.

	the total	how much stuff (yours)	the total ÷ the stuff (the place's)
At this store	$15	12 ounces	$1.25 per ounce
Gravitational	W in joules	m in kilograms	U in J/kg
Electrostatic	W in joules	q in coulombs	V in J/C, the volt

the last column is the FIELD or the POTENTIAL — the price per unit of stuff

and the same grid with newtons in the second column gives g in N/kg and E in N/C

Figure 1. The same three columns every time. **The quantity you are usually handed is in the first two columns; the one physics cares about is in the third.** Note what is *not* in the third column: nothing about the particular bag, mass or charge you happened to bring. The third column is the store's, not yours — and that is the whole point of a field.

For electrostatic, the potential is commonly known as **voltage**, **volts**, or the **potential difference** with its own unit, V (J/C). So, imagine you have 3 C ("3 ounces"). This charge is in the potential of 2 V ("the unit price"). What is the total energy ("the total dollar amount")?

Practice

1. A charge has an energy of 3 J (the total dollar amount), which is in the potential of 0.5 V (the unit price or price per ounce). How much of a charge was in the potential (the amount in ounces)?

2. A charge is sitting in the potential of 2 V. This charge has the energy of 10 J. How much charge is in the potential?

3. A charge of 3 C (the amount in ounces) is in the electric field of 3 N/C (the unit price). What force in newtons (N) does this charge experience (the total dollar amount)?

4. A charge of 2 C is experiencing a force of 10 N. What is the strength of the electric field (N/C)?

Why one of them is harder than the other

The supermarket stops being useful here, and it is worth saying so out loud rather than letting it quietly mislead you: **a price has no direction.** Everything from this point on turns on the difference between a quantity that has one and a quantity that does not — so put the coffee down.

Let's talk a bit more about the differences between field (force-like) and potential (energy-like).

Forces are vectors, and fields are force-like; the fields add like vectors, meaning you need to care about the **direction** of the field. This direction (as well as the magnitude) is revealed by placing a **test charge** (+1 unit charge). Let's draw the fields of two charges and see how they add.

Energies are scalars, and potentials are energy-like; the energy adds like scalars (like numbers), meaning they just add **with signs** (positive potential for positive charge and negative potential for negative charge). Let's draw the potentials of two charges and see how they add.

DRAW THEM FIRST

Get graph paper and do this before you read on. Put two charges on a line. Along that line, plot what **each** charge does on its own, then add them — **as arrows for the field, as numbers for the potential**. Do it for all three cases: two positives, two negatives, and one of each.

Then compare with the plots you have already seen: **Note 3, Figure 6** for the field and **Note 4, Figure 3** for the potential. Those are these same three cases, drawn from the same sheets.

Line charges, and a field that stops changing

Finally, the line charges. Why do you make line charges? See the pictures in "Electric force near a sheet of charge." When you line up charges to make sheets, **the field becomes constant between the sheets** (or lines). This constant electric field is similar to **the gravitational field near the surface of the Earth**.

That is the single most useful thing in this module, and it is why $g = 9.8\ \text{m/s}^2$ works at all: close to something big and flat, the field stops caring where exactly you are. **Note 4, Figure 4** is the picture — the lines run straight and evenly spaced between the two rows, and sag only at the ends.

Equipotential, and the contour map

Flip over, and we will now draw the potential. The gravitational potential varies with **the height you are from the surface**. Remember, this is the same as the potential energy going up with the height. So, **the potential is the same where the height is the same**. The locations where the potential is the same is called **equipotential**. This is akin to **a contour map that shows the lines of equal altitudes.**



Figure 2. His analogy, drawn. The dashed rules are the four altitudes, and each ring on the right is drawn at the radius where the hill on the left **actually** reaches that height — the rings are computed from the profile, not spaced by eye. **Crowded rings mean a steep hill**; on an equipotential map crowded lines mean a strong field, which is the same sentence with two words changed.

Note 4, Figure 2 is this map for two real charges, and **Note 5** turns the picture into arithmetic: the steeper the hill, the bigger the field.

THE PUNCHLINE, IN HIS WORDS

the price per unit “stuff”

Force and energy are what you are handed. **Field and potential are what you get when you divide out the stuff** — and because the stuff is gone, what is left belongs to the space itself rather than to whatever you happened to put in it. That is why you can draw a field map of an empty room.

Looking ahead

You have the meaning of both quantities now, and a picture for each. **Note 12** spends all of it on real problems — alpha decay, how hard fusion is, and a pair of plates used as an accelerator — all of which are bookkeeping between U and K once you have the potentials this module has built.